SPECIFIC AIMS

Oropharyngeal cancer (OPC) caused by oncogenic human papillomavirus (HPV) is increasing rapidly in incidence in the United States general population and at an even more rapid pace among Veterans. HPV vaccination prevents chronic oral HPV infection, the precursor to HPV+ OPC, and is universally recommended up to age 26 in males and females. There are also recommendations for vaccination up to age 45 through shared decision-making, primarily in those with high-risk patients (MSM, transgender persons, and those with immunocompromising conditions). There is significant controversy and conflicting recommendations regarding the benefit of universal HPV vaccination for individuals between 26 and 45 years of age, with several recent studies suggesting a lack of public health benefits or cost-effectiveness in the general population. However, we do not believe that simply extrapolating these data to the Veteran population is appropriate since 1) previous work demonstrating a lack of public health benefit focuses primarily on cervical cancer and the vast majority of Veterans are male, 2) the risk of HPV+ OPC is increasing disproportionately among Veterans, 3) unlike cervical cancer, there is no effective screening for HPV+ OPC, and 4) the onset of HPV+OPC occurs on average 15-20 years later than cervical cancer, 5) military-related exposures may increase head and neck cancer risk (i.e. burn pits). Thus, there is an urgent need to investigate the potential benefit of HPV vaccination among Veterans up to age 45 that accounts for the unique characteristics of the VHA population. The purpose of this study is to assess the cost-effectiveness and potential public health benefit of universal HPV vaccination in Veterans up to age 45, as well as the first qualitative assessment of Veteran attitudes and perceptions towards HPV vaccination. Completion of the aims detailed below should generate meaningful data on the current state of HPV vaccination within the VHA and could help define future policy recommendations that save Veteran lives and reduce the financial burden of this deadly disease. The specific aims of this proposal are to:

Aim 1. Determine the overall prevalence of HPV vaccinated Veterans aged 26-45 and measure current trends in HPV vaccine administration by the VHA in this age group: Using data from the VHA Corporate Data Warehouse (CDW), we will estimate the prevalence of HPV vaccination in Veterans age 26-45 and examine vaccination rates by race/ethnicity, socioeconomic status, rurality based on Rural-Urban Commuting Area Codes (RUCAs), and geographical location based on Veteran Integrated Service Network (VISN). All Veterans with at least one primary care visit from 2018 to 2023 will be included in this analysis. We will also measure trends in HPV vaccine administration by the VHA since the ACIP recommendations were instituted in 2018. We hypothesize that HPV vaccination rates in this age group are increasing through shared decision-making.

Aim 2: Model the potential public health benefit and cost-effectiveness of extending the age of universal HPV vaccination eligibility in the VHA population. While there is controversy regarding the role of HPV vaccination beyond age 26 in the general population, the public health benefit in the Veteran population has not been investigated. We will conduct an HPV vaccination simulation model to assess the benefits and costs of extending HPV vaccination to age 30, 35, 40, and 45 years with the VHA. We hypothesize that there will be a greater public health benefit compared to the general population due to a predominantly male population, later onset and a higher incidence of HPV-associated oropharyngeal cancer compared to other HPV-associated cancers, and a lack of effective screening.

Aim 3: Explore attitudes and perceptions towards HPV vaccination in Veterans age 26 to 45 years of age and VHA primary care providers: We will conduct a convergent parallel mixed methods study that combines structured questionnaires, focus group discussions, and in-depth interviews to, a) explore knowledge and awareness of HPV vaccination among Veterans age 26-45, b) assess perceptions, beliefs, misconceptions, and concerns about HPV vaccination, and c) identify barriers and facilitators that influence vaccine acceptance or hesitancy among this age group. Veterans from diverse demographic, geographic, and military service backgrounds will be recruited for this study. The results of this study can be used to design interventions and make policy recommendations on HPV vaccination specifically tailored to Veterans.